Hessian_Matrix

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06 February 2019

All of the results that follow assume continuity of all of the derivatives.

If f is a function of n variables, with vector of inputs $\mathbf{x} = (x_1, x_2, ..., x_n)$ and scalar output $f(\mathbf{x})$, then the **Hessian Matrix H** of f is the matrix of it's second partial derivatives:

$$\mathbf{H} \quad = \quad \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ . & . & . & . \\ . & . & . & . \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Example

Consider the function

$$f(\mathbf{x}) \quad = \quad f(x, y, z) \quad = \quad x^2 e^y + \frac{5xz^3}{y}$$

The hessian matrix for this function is:

$$\mathbf{H} \quad = \quad \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial y \partial z} \\ \frac{\partial^2 f}{\partial z \partial x} & \frac{\partial^2 f}{\partial z \partial y} & \frac{\partial^2 f}{\partial z^2} \end{bmatrix}$$

The second partial derivatives are:

$$\frac{\partial^2 f}{\partial x^2} \quad = \quad \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) \quad = \quad \frac{\partial}{\partial x}\left(2xe^y + \frac{5z^3}{y}\right) \quad = \quad 2e^y$$

$$\frac{\partial^2 f}{\partial x \partial y} \quad = \quad \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) \quad = \quad \frac{\partial}{\partial x}\left(x^2 e^y - 5xz^3 y^{-2}\right) \quad = \quad 2xe^y - \frac{5z^3}{y^2}$$

$$\frac{\partial^2 f}{\partial x \partial z} \quad = \quad \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial z}\right) \quad = \quad \frac{\partial}{\partial x}\left(\frac{15xz^2}{y}\right) \quad = \quad \frac{15z^2}{y}$$

$$\frac{\partial^2 f}{\partial y \partial x} \quad = \quad \frac{\partial^2 f}{\partial x \partial y} \qquad \text{Schwarz/Clairaut's Theorem}$$

$$\frac{\partial^2 f}{\partial y^2} \quad = \quad \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) \quad = \quad \frac{\partial}{\partial x}\left(x^2 e^y - \frac{5xz^3}{y^2}\right) \quad = \quad x^2 e^y + \frac{10xz^3}{y^3}$$

$$\frac{\partial^2 f}{\partial y \partial z} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial z}\right) = \frac{\partial}{\partial y}\left(\frac{15xz^2}{y}\right) = -\frac{15xz^2}{y^2}$$

$$\frac{\partial^2 f}{\partial z \partial x} = \frac{\partial^2 f}{\partial x \partial z}$$

$$\frac{\partial^2 f}{\partial z \partial y} = \frac{\partial^2 f}{\partial y \partial z}$$

$$\frac{\partial^2 f}{\partial z^2} = \frac{\partial}{\partial z}\left(\frac{\partial f}{\partial z}\right) = \frac{\partial}{\partial z}\left(\frac{15xz^2}{y}\right) = \frac{30xz}{y}$$

So, the **Hessian matrix** for $f(x, y, z)$ is

$$\mathbf{H} = \begin{bmatrix} 2e^y & 2xe^y - \frac{5z^3}{y^2} & \frac{15z^2}{y} \\ 2xe^y - \frac{5z^3}{y^2} & x^2e^y + \frac{10xz^3}{y^3} & -\frac{15xz^2}{y^2} \\ \frac{15z^2}{y} & -\frac{15xz^2}{y^2} & \frac{30xz}{y} \end{bmatrix}$$

For example, the Hessian matrix at the point $\mathbf{x} = (1, 2, 3)$ is

```
x <- 1
y <- 2
z <- 3
matrix( c(2*exp(y),                  2*x*exp(y)-5*z^3/y^2,      15*z^2/y,
          2*x*exp(y)-5*z^3/y^2,      x^2*exp(y)+10*x*z^3/y^3,   -15*x*z^2/y^2,
          15*z^2/y,                  -15*x*z^2/y^2,             30*x*z/y
         ),
        nrow=3, ncol=3, byrow=3
      )
```

```
##           [,1]      [,2]   [,3]
## [1,]  14.77811 -18.97189  67.50
## [2,] -18.97189  41.13906 -33.75
## [3,]  67.50000 -33.75000  45.00
```

```
rm(x);rm(y);rm(z)
```

and here is a numerical approximation of the Hessian matrix in R (to validate my calculations):

```
library(pracma)
```

```
## Warning: package 'pracma' was built under R version 3.5.1
```

```
function_xyz <- function(x){ x[1]^2*exp(x[2]) + 5*x[1]*x[3]^3 / x[2] }

pracma::hessian( f = function_xyz,
                 x0 = c(1,2,3)
                )
```

```
##           [,1]      [,2]   [,3]
## [1,]  14.77811 -18.97189  67.50
## [2,] -18.97189  41.13906 -33.75
## [3,]  67.50000 -33.75000  45.00
```